

UNIT I

FOUNDATION OF LEARNING

Types of Learning - **Supervised, Unsupervised, Reinforcement**

Components of learning

Learning Vs. Design

Learning Models - **Geometric, probabilistic, logical**

Theory of Learning

Feasibility of Learning

Grouping and Grading, Error and noise, Bias and Variance

Generalization - **Theory of Generalization, Generalization bound,**

Approximation generalization trade off

TYPES OF LEARNING SUPERVISED LEARNING

Supervised learning involves learning a function from input-output pairs. The algorithm is trained on a labeled dataset, meaning each input has a known output.

- Classification (e.g., spam detection)
- Regression (e.g., house price prediction)

TYPES OF LEARNING UNSUPERVISED LEARNING

Unsupervised learning finds patterns in data without labels. It identifies structure or groups in the input data.

- Clustering (e.g., customer segmentation)
- Dimensionality reduction (e.g., PCA)

TYPES OF LEARNING REINFORCEMENT LEARNING

In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties.

- Positive Reinforcement
- Negative Reinforcement

COMPONENTS OF LEARNING

- 1) Feature Extraction
- 2) Feature Selection
- 3) Model Selection
- 4) Training
- 5) Evaluation
- 6) Testing

LEARNING VS DESIGN

Aspect	Learning-Based Approach	Design-Based Approach
Definition	System learns patterns from data	System behavior is manually programmed
Core Mechanism	Learns a function/model from experience	Uses predefined rules and logic
Adaptability	High – adapts to new/unseen data	Low – fixed behavior, no adaptation
Data Requirement	Requires large datasets for training	No training data needed
Development Time	Slower initial development (due to data/training)	Faster initial setup
Accuracy	Can improve with more data and better models	Limited to designer's understanding
Interpretability	Often a "black box" (hard to interpret)	Transparent and easy to trace
Scalability	Scales well with complex and large data	Hard to scale for complex problems
Error Handling	Learns from errors and improves	Must be manually debugged and updated
Example Use Case	Email spam filter using Naive Bayes	Email spam filter using keyword-based rules
Example Field	Image recognition, NLP, recommendation systems	Form validation, simple control systems
Performance on Noisy Data	Better if trained properly	Struggles unless explicitly handled
Human Expertise	Requires domain knowledge to preprocess data	Requires domain knowledge to write rules

LEARNING MODELS

Model Type	Based On	Examples
Geometric	Geometry, distances, hyperplanes	k-NN, SVM, Perceptron
Probabilistic	Probability theory	Naive Bayes, Logistic Regression
Logical	Logic and rule-based reasoning	Decision Trees, Rule-based learners

GROUPING AND GRADING

- Grouping: Process of organizing data into clusters or categories based on similarity.
- Grading: Assigning a quality, level, or rank to data points, model predictions, or performance metrics.

NOISE AND ERROR

- Error: Difference between predicted output and true output.

$$\text{Error} = y - y^{\wedge}$$

- Noise: Irreducible randomness or error in data that cannot be explained or learned by any model.

BIAS AND VARIANCE

- Bias: Error due to simplified assumptions in the model.

High bias → Underfitting

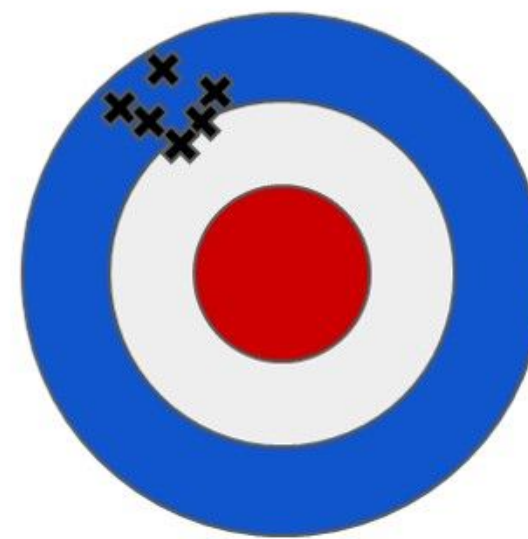
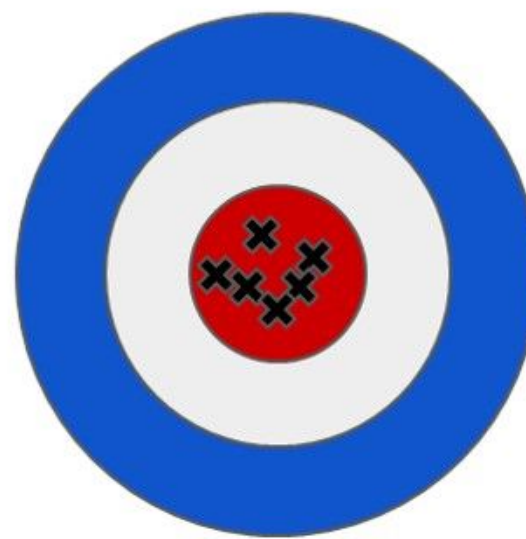
- Variance: Error due to model sensitivity to training data.

High variance → Overfitting

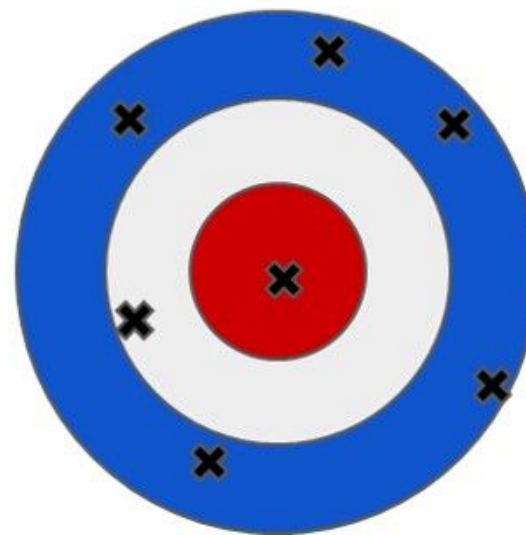
Low bias

High bias

Low variance



High variance



Learning Curve

